

Disproof of Conjecture 00000007608: a rotation has infinitely many decompositions into two reflections

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Abstract

Conjecture 00000007608 asserts that the composition of two reflections is a rotation, that the decomposition (of a rotation into two reflections) is unique, and that the exception to uniqueness is the case of parallel reflections. The uniqueness clause is false for *every* rotation $\rho \neq \text{id}$ of the plane. Writing R_α for the reflection in the line through the centre at angle α , we have $\rho_\theta = R_\beta \circ R_\alpha$ whenever $\beta - \alpha \equiv \theta/2 \pmod{\pi}$, so α can be chosen freely. None of these decompositions has parallel mirrors. The simplest instance is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Its four mirrors are the x -axis, the line $y = x$, the y -axis and the line $y = -x$. In Lean 4 (core library only) we formalise rational 2×2 reflections and rotations from scratch. We prove that the decompositions of any rational rotation ρ are exactly the pairs $(A, \rho A)$. We prove that for every rational $\rho \neq \text{id}$ the Pythagorean family A_t ($t \in \mathbb{N}$) gives infinitely many non-parallel decompositions with pairwise different mirrors, no three of which share an unordered mirror pair. The uniqueness clause is refuted in its weakest literal form (unordered pairs of mirrors). We also treat $\mathbb{R}^3$ and affine reflections, and we discuss the one charitable reading under which the clause becomes true: “unique up to rotating both mirrors” and “given one mirror, the other is unique”.

1 The claim

The English statement reads: “*Definition: Reflections as hyperplane mirror structure. Conjecture: The composition of two reflections is a rotation and the decomposition is unique; the exception of uniqueness is parallel reflections; and the exception of parallel reflections is translation composition.*” The Chinese version says the same thing: the composition of two reflections is a rotation and the decomposition is unique (“fen jie wei yi”); the exception to uniqueness is parallel reflections; the exception for parallel reflections is a composite translation. We split it into four clauses:

- (C1) the composition of two reflections is a rotation;
- (C2) the decomposition of a rotation into two reflections is unique;
- (C3) the exception to (C2) is the case of parallel mirrors;
- (C4) the exception to (C1) is the case of parallel mirrors, whose composition is a translation.

Clauses (C2) and (C3) together form the *uniqueness clause*:

- (U) If ρ is a rotation and $\rho = B \circ A = B' \circ A'$ with reflections A, B, A', B' , and neither pair has parallel mirrors, then the two decompositions coincide.

We refute (U). Because the conjecture is a conjunction, this refutes the conjecture.

Convention. The disproof relies on the literal reading of the bilingual text. The Chinese text says that the decomposition is unique and that the exception to uniqueness is parallel reflections (“fen jie wei yi; qie wei yi xing de li wai wei ping xing fan she”); the English text says the same. Clauses (C1) and (C4) are correct. (C1) holds for linear reflections, and for affine reflections it fails exactly for parallel distinct mirrors, which is the exception named in (C4); see Section 5. So the conjunction fails through the uniqueness clause (U) and only through it. Section 5 discusses two weaker, true readings of (U) that we do not refute.

2 Set-up

A *reflection* of $\mathbb{R}^2$ (fixing the origin) is an orthogonal matrix A with $\det A = -1$ and $A^2 = I$. A *rotation* is an orthogonal matrix with $\det = +1$. The *mirror* of a reflection A is its fixed line $\text{Fix } A = \ker(A - I)$. Two mirrors through the origin are parallel iff they are equal. For $\alpha, \theta \in \mathbb{R}$ put

$$R_\alpha = \begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix}, \quad \rho_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

R_α is the reflection in the line $\mathbb{R}(\cos \alpha, \sin \alpha)$, and $R_\alpha = R_{\alpha'}$ iff $\alpha \equiv \alpha' \pmod{\pi}$.

Lemma 1. *Over any ordered field K (in particular $\mathbb{Q}$ or $\mathbb{R}$), an orthogonal 2×2 matrix with determinant -1 has the form $\begin{pmatrix} a & b \\ c & -a \end{pmatrix}$ with $a^2 + b^2 = 1$, and it is automatically an involution. An orthogonal matrix with determinant $+1$ has the form $\begin{pmatrix} x & -y \\ y & x \end{pmatrix}$ with $x^2 + y^2 = 1$. Over $\mathbb{R}$ these are exactly the R_α and the ρ_θ .*

Proof. Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $MM^T = I$, so $a^2 + b^2 = c^2 + d^2 = 1$. If $ad - bc = -1$, then $(b - c)^2 + (a + d)^2 = (a^2 + b^2) + (c^2 + d^2) + 2(ad - bc) = 0$, so $c = b$ and $d = -a$. Then $M^2 = (a^2 + b^2)I = I$. If $ad - bc = 1$, then $(b + c)^2 + (a - d)^2 = 2 - 2 = 0$ gives $c = -b$ and $d = a$. $\square$

Lemma 2. $R_\beta R_\alpha = \rho_{2(\beta - \alpha)}$ and $\rho_\theta R_\alpha = R_{\alpha + \theta/2}$.

Proof. By direct multiplication and the addition formulas,

$$R_\beta R_\alpha = \begin{pmatrix} \cos 2\beta \cos 2\alpha + \sin 2\beta \sin 2\alpha & \cos 2\beta \sin 2\alpha - \sin 2\beta \cos 2\alpha \\ \sin 2\beta \cos 2\alpha - \cos 2\beta \sin 2\alpha & \sin 2\beta \sin 2\alpha + \cos 2\beta \cos 2\alpha \end{pmatrix} = \rho_{2\beta - 2\alpha}.$$

The second identity follows by multiplying the first, with $\beta = \alpha + \theta/2$, on the right by $R_\alpha = R_\alpha^{-1}$. $\square$

3 The main theorem

Theorem 3. *Let $\rho = \rho_\theta \neq I$ be a rotation of $\mathbb{R}^2$.*

1. *The decompositions $\rho = B \circ A$ into two reflections are exactly the pairs $(A, \rho A)$ with A an arbitrary reflection, i.e. $(R_\alpha, R_{\alpha + \theta/2})$ for $\alpha \in \mathbb{R}/\pi\mathbb{Z}$.*
2. *In none of them are the two mirrors parallel.*
3. *Distinct $\alpha \in \mathbb{R}/\pi\mathbb{Z}$ give distinct ordered pairs and distinct first mirrors. Each unordered pair of mirrors arises for at most two values of α . Hence there are uncountably many decompositions, even counted as unordered pairs of mirrors.*

Proof. (1) If $BA = \rho$, then $B = BAA = \rho A$. Conversely, if $A = R_\alpha$, then $\rho A = R_{\alpha+\theta/2}$ is a reflection (Lemma 2), and $(\rho A)A = \rho A^2 = \rho$. (2) If A and $B = \rho A$ had the same mirror, a nonzero vector v on it would satisfy $Av = v = Bv = \rho Av = \rho v$. A rotation $\neq I$ fixes no nonzero vector, since $\det(\rho_\theta - I) = 2 - 2\cos\theta \neq 0$. Equivalently, the mirrors are at angle $\theta/2 \not\equiv 0 \pmod{\pi}$. (3) The first mirror determines α modulo π . An unordered pair $\{m, m'\}$ contains at most two lines, so at most two decompositions have their first mirror in it. $\square$

So uniqueness fails for *every* nontrivial rotation. The exception named in (C3), parallel mirrors, never occurs for $\rho \neq I$: parallel (equal) linear mirrors give $A = B$ and $BA = I$. The only rotation that has decompositions with parallel mirrors is the identity, and for it *all* decompositions (A, A) have equal mirrors.

4 The rational version (formalised in Lean)

Lean's core library has no real numbers, so the formal proof works over $\mathbb{Q}$. That is enough, because the counterexamples are rational. A rational matrix is stored as m/q , with m an integer matrix and $q > 0$, and all predicates are the rational ones with the denominators cleared. Lemma 1 holds verbatim over $\mathbb{Q}$, where the last step uses that a sum of two rational squares vanishes only if both do. It is formalised as `isRefl_iff`, `rot_form`, `refl_of_form` and `rot_of_form`.

Proposition 4 (general statements, all rational rotations). *Let ρ be a rational rotation and A, B rational reflections.*

1. BA is a rotation (`refl_mul_refl`; this is (C1) in the linear case).
2. ρA is a reflection and $(\rho A)A = \rho$ (`rot_mul_refl`, `decomp_of_refl`).
3. If $BA = \rho$, then $B = \rho A$ (`decomp_classification`).

Proof. Write $\rho = \begin{pmatrix} x & -y \\ y & x \end{pmatrix}$ with $x^2 + y^2 = 1$ and $A = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}$ with $a^2 + b^2 = 1$. Then $\rho A = \begin{pmatrix} a' & b' \\ b' & -a' \end{pmatrix}$ with $a' = xa - yb$ and $b' = xb + ya$. Moreover $a'^2 + b'^2 = (x^2 + y^2)(a^2 + b^2) = 1$, so ρA is a reflection by Lemma 1. Also $(\rho A)A = \rho(a^2 + b^2) = \rho$. Parts (1) and (3) are similar polynomial identities, given in the Lean file with the denominators cleared. $\square$

The Pythagorean family. For $t \in \mathbb{N}$ let

$$A_t = \frac{1}{1+t^2} \begin{pmatrix} 1-t^2 & 2t \\ 2t & t^2-1 \end{pmatrix}.$$

Since $(1-t^2)^2 + (2t)^2 = (1+t^2)^2$, Lemma 1 shows that A_t is a reflection (`Aref_refl`). Moreover $A_t(1, t)^T = (1, t)^T$ (`Aref_fix`), so the mirror of A_t is the line of slope t . If A_s fixes $(1, t)$, then the second coordinate gives $2s + (s^2 - 1)t = (1 + s^2)t$, i.e. $s = t$ (`Aref_fix_eq`). Hence the mirrors of the A_t are pairwise different (`Aref_distinct`), and so are the matrices (`Aref_injective`). Examples: $A_0 = \text{diag}(1, -1)$, $A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $A_2 = \frac{1}{5} \begin{pmatrix} -3 & 4 \\ 4 & 3 \end{pmatrix}$.

Lemma 5 (`nonparallel`). *If ρ is a rational rotation with $\rho \neq I$, then A_t and ρA_t have different mirrors.*

Proof. Otherwise ρA_t fixes $v = (1, t)$. Since $A_t v = v$, this gives $\rho v = v$, that is, $x - yt = 1$ and $y + xt = t$. Multiplying the first equation by t and subtracting the second gives $y(1 + t^2) = 0$, so $y = 0$ and $x = 1$, i.e. $\rho = I$. $\square$

Theorem 6 (`infinitely_many_decompositions, not_uniqueFor`). *Let $\rho \neq I$ be a rational rotation, and let $D_t = (A_t, \rho A_t)$ for $t \in \mathbb{N}$. Then:*

1. *every D_t is a decomposition $\rho = (\rho A_t) \circ A_t$ with non-parallel mirrors;*
2. *for $s \neq t$ the first mirrors of D_s and D_t differ;*
3. *for pairwise distinct s, t, u , the unordered mirror pairs of D_s, D_t, D_u are not all equal.*

In particular (U) fails for ρ , even when decompositions are compared as unordered pairs of mirrors.

Proof. (1) is Proposition 4(2) together with Lemma 5. (2) is `Aref_distinct`. For (3), suppose all three unordered pairs coincide. Write m_t for the mirror of A_t and m'_t for that of ρA_t . Since $m_t \neq m_s$, equality of $\{m_s, m'_s\}$ and $\{m_t, m'_t\}$ forces $m_t = m'_s$; likewise $m_u = m'_s$. Hence $m_t = m_u$, contradicting (2). Applying (U) to D_0, D_1 and to D_0, D_2 gives exactly such a coincidence. $\square$

Corollary 7 (`conjecture_00000007608_false`). *(U) is false. It already fails for the rotation by 90° , $\rho = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.*

Explicit integer examples. Let $R_1 = \text{diag}(1, -1)$, $R_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $R_3 = \text{diag}(-1, 1)$ and $R_4 = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$. Their mirrors are the x -axis, the line $y = x$, the y -axis and the line $y = -x$; these are pairwise different (`four_mirrors_distinct`, witnessed by the vectors $(1, 0)$, $(1, 1)$ and $(0, 1)$). Then:

- $R_2 R_1 = R_4 R_3 = \rho_{90^\circ}$ (`rot90_two_decompositions`);
- $R_3 R_1 = R_4 R_2 = R_2 R_4 = -I = \rho_{180^\circ}$ (`rot180_two_decompositions`);
- $\rho_{90^\circ} A_2 = \frac{1}{5} \begin{pmatrix} -4 & -3 \\ -3 & 4 \end{pmatrix}$ (`Aref2_example`).

5 Readings

- **Ordered vs. unordered.** Lean refutes the weakest literal form, `UniqueClause`: two non-parallel decompositions should have the same *unordered* pair of mirrors. The ordered, matrix-level form `UniqueClauseOrdered` implies it (`ordered_implies_unordered`), so it is false too (`conjecture_false_ordered`).
- **Which rotation.** Non-uniqueness is not a special phenomenon of one rotation. It holds for every rotation $\neq I$: Theorem 3 over $\mathbb{R}$, and `every_nontrivial_rotation_fails` for all rational rotations in Lean. For $\rho = I$ every decomposition is (A, A) , with equal (“parallel”) mirrors, so (U) holds vacuously there (`uniqueFor_id`). This also shows that the formal predicate is not trivially false.
- **Real vs. rational.** Lean works over $\mathbb{Q}$. Theorem 3 is the real statement, and `verify.py` illustrates it numerically for irrational angles.
- **Affine reflections.** An affine reflection is $v \mapsto Mv + u$, with M a linear reflection, that is an involution; its mirror is an affine line. A rotation about a point c is $\rho_c(v) = \rho(v - c) + c$. If two mirrors meet in a point P , the composition fixes P . If they are parallel and distinct, the composition is a nonzero translation. For example, the mirrors $x = 0$ and $x = 1$ give $v \mapsto v + (2, 0)$, which has no fixed point (`parallel_gives_translation`). So for $\rho_c \neq \text{id}$ both mirrors pass through c , and conjugating by the translation $v \mapsto v - c$ identifies the affine

decompositions of ρ_c with the linear decompositions of ρ . The affine reading therefore has the same infinite family. Concretely, the rotation by 90° about $(1, 1)$ equals $S_{y=x} \circ S_{y=1} = S_{x+y=2} \circ S_{x=1}$, with four different non-parallel mirrors through $(1, 1)$ (`affine_two_decompositions`).

- **$\mathbb{R}^n$ and hyperplane mirrors.** The definition speaks of hyperplane mirrors. In $\mathbb{R}^n$ a product of two hyperplane reflections with distinct mirrors H_1, H_2 is a rotation in the plane $(H_1 \cap H_2)^\perp$ about the axis $H_1 \cap H_2$. A rotation $\rho \neq I$ of this kind (angle θ in a plane Π , fixing $\Pi^\perp$) is decomposed exactly by the pairs of hyperplanes containing $\Pi^\perp$, i.e. $H_i = \ell_i \oplus \Pi^\perp$, where (ℓ_1, ℓ_2) runs over the decompositions of $\rho|_\Pi$ in Theorem 3. Again there is a circle's worth. In Lean, `r3_two_decompositions` embeds the 90° example block-diagonally in $\mathbb{R}^3$. There the mirrors are the planes $y = 0$, $x = y$, $x = 0$ and $x = -y$, each containing the z -axis, and they are pairwise different. Parallel hyperplanes through the origin are equal, so the exception again never applies.
- **Point reflections.** If “reflection” meant a point reflection $v \mapsto 2c - v$, the composition of two of them would be a translation, never a nontrivial rotation, and (C1) would fail outright. The definition (“hyperplane mirror structure”) excludes this reading.
- **Charitable reading 1: “unique up to rotating both mirrors”.** Standard texts state the *non*-uniqueness: a rotation about c is a product of two reflections in lines through c making half the angle of rotation, and one of the two mirrors may be chosen arbitrarily through c [1, 2]. What is determined is only the angle between the mirrors, $\theta/2 \bmod \pi$, so the pair is *unique up to rotating both mirrors simultaneously about the centre*. This weakened statement is true and follows from Theorem 3(1): the decompositions $(R_\alpha, R_{\alpha+\theta/2})$ and $(R_{\alpha'}, R_{\alpha'+\theta/2})$ differ by rotating both mirrors by $\alpha' - \alpha$. We do *not* refute this version. But it is not what the conjecture states, and the standard sources, which stress that a mirror can be chosen freely, support reading “unique” literally as uniqueness of the pair of mirrors. Under the “up to rotation” reading there would also be no exception at all for $\rho \neq I$, and parallel mirrors would be a different case, not an exception to uniqueness.
- **Charitable reading 2: relative uniqueness.** “Given one of the two mirrors, the other is unique.” This is also true. It is exactly Theorem 3(1), and in Lean it is `decomp_classification`: $BA = \rho$ forces $B = \rho A$, and symmetrically $A = B\rho$. Again it is not what is stated. Relative uniqueness holds equally for translations: given one of two parallel mirrors, the other is determined by the translation vector. So under this reading the stated exception (parallel mirrors) would be vacuous, which the conjecture plainly does not intend.
- **Clause (C1).** For linear reflections (C1) is true: BA is orthogonal with $\det = (-1)^2 = 1$ (`refl_mul_refl`). With equal mirrors the composition is the identity, the rotation by 0. For affine reflections with parallel distinct mirrors, (C1) is false: the composition is a translation with no fixed point. This is the exception that (C4) mentions, so (C1) and (C4) together are correct. The disproof does not use them.

References

- [1] H. S. M. Coxeter, *Introduction to Geometry*, 2nd ed., Wiley, 1969, Chapter 3 (isometries of the Euclidean plane; a rotation as a product of two reflections).

- [2] D. A. Brannan, M. F. Esplen, J. J. Gray, *Geometry*, 2nd ed., Cambridge University Press, 2012 (Euclidean geometry; isometries as products of reflections).

6 Machine verification

Lean 4 (`lean4/Main.lean`, **core only**, **v4.19.0**). From scratch it defines:

- integer 2×2 matrices `M2` (product, transpose, determinant, action on $\mathbb{Z}^2$), and rational matrices `QM` = m/q with equality `QEq`, proved to be an equivalence relation;
- `IsRefl` (orthogonal, $\det = -1$, involution), `IsRot`, the fixed-vector predicate `Fix`, `SameMirror` (parallel = equal mirrors), and `Decomp`;
- the claims `UniqueFor  $\rho$` , `UniqueClause` and `UniqueClauseOrdered`.

The main theorems are:

- `conjecture_00000007608_false` : $\neg$ `UniqueClause`;
- `conjecture_false_rot90_explicit` (the same, directly from $R_2 R_1 = R_4 R_3$);
- `conjecture_false_ordered`;
- `every_nontrivial_rotation_fails`;
- `infinitely_many_decompositions`;
- `decomp_classification`, `refl_mul_refl`, `isRefl_iff` and `uniqueFor_id`;
- the concrete 90° , 180° , $\mathbb{R}^3$ and affine examples.

A small tactic `ring.z` (expand, AC-normalise, `omega`) proves the polynomial identities, since core Lean has no `ring`. The concrete checks use `decide`. There is no `sorry` and no `native_decide`. The axioms used are at most `propext` and `Quot.sound`.

Limitations. Lean works over $\mathbb{Q}$, not $\mathbb{R}$; the real statement (Theorem 3) is proved in this report. Mirrors are compared through integer vectors, which loses nothing, since a rational vector is fixed iff an integer multiple of it is. The $\mathbb{R}^3$ and affine parts are concrete integer examples (`decide`). The general $\mathbb{R}^n$ and affine reductions above, and the “up to rotation” statement, are proved only in this report.

Python (`verify.py`, **standard library**). A different method: exact `Fraction` matrices, mirrors computed as kernels by Gaussian elimination, a small polynomial engine for the symbolic identities, and brute force. For 123 nontrivial rational rotations and 86 rational parameters t , every $(A_t, \rho A_t)$ is a non-parallel decomposition with pairwise different mirrors. In a set S of 124 rational reflections it enumerates all pairs $(A, B) \in S^2$ with $BA = \rho$. For ρ_{90° there are 124 such pairs, all non-parallel, with 124 different unordered mirror pairs. For $\rho = I$ all of them are (A, A) . It also checks the $\mathbb{R}^3$ and affine examples, and evaluates $R_{\alpha+\theta/2} R_\alpha = \rho_\theta$ in floating point for irrational angles.